

## ● RAG Learning

## B6.3 Monitoring & Maintaining Health

OCR A GCSE Biology (J247) Gateway — Topic B6: Global challenges

### 1. Health and Disease

The WHO defines **health** as a state of complete physical, mental and social wellbeing — not just the absence of disease. **Disease** = a condition that affects normal body function.

Diseases fall into two broad categories: **communicable** (can be passed from one organism to another) and **non-communicable** (cannot be passed on).

| Communicable                                             | Non-communicable                                  |
|----------------------------------------------------------|---------------------------------------------------|
| Caused by pathogens (bacteria, viruses, fungi, protists) | Caused by lifestyle, genetics or environment      |
| e.g. flu, measles, TB, malaria, athlete's foot           | e.g. heart disease, type 2 diabetes, most cancers |

### 2. Pathogens and Communicable Disease

| Pathogen type   | Examples                                 | How it spreads                                                                 |
|-----------------|------------------------------------------|--------------------------------------------------------------------------------|
| <b>Bacteria</b> | Cholera, TB, food poisoning (Salmonella) | Multiply rapidly; release toxins that damage cells/tissues                     |
| <b>Viruses</b>  | Flu, measles, HIV, COVID-19              | Reproduce inside host cells; eventually burst the cell, releasing more viruses |

|                                            |                                 |                                                             |
|--------------------------------------------|---------------------------------|-------------------------------------------------------------|
| <b>Fungi</b>                               | Athlete's foot, ringworm        | Hyphae grow into skin; release digestive enzymes            |
| <b>Protists</b> (single-celled eukaryotes) | Malaria (carried by mosquitoes) | Often need a vector (e.g. mosquito) to spread between hosts |

## Methods of disease transmission

- **Direct contact** — touching (e.g. athlete's foot, STIs)
- **Air** — droplets from sneezes/coughs (e.g. flu, COVID-19, TB)
- **Water** — contaminated water (e.g. cholera)
- **Food** — contaminated food (e.g. Salmonella, E. coli)
- **Vector** — animal carrier (e.g. mosquitoes carry malaria)
- **Body fluids** — blood, sex (e.g. HIV)

## Reducing transmission

- Hand washing, hygiene
- Cooking food properly; clean water supply
- Sterilising medical instruments
- Isolating infected people
- Vector control (e.g. insecticides, bed nets for malaria)
- Vaccination (see below)

## 3. Body Defences Against Pathogens

### Non-specific (first-line) defences

| Barrier     | How it works                                                               |
|-------------|----------------------------------------------------------------------------|
| <b>Skin</b> | Physical barrier; produces antimicrobial chemicals; scab forms over wounds |

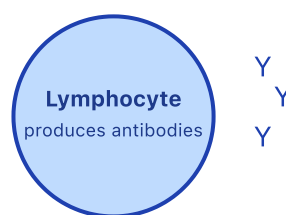
|                                   |                                                                                            |
|-----------------------------------|--------------------------------------------------------------------------------------------|
| <b>Mucus and cilia in airways</b> | Mucus traps pathogens; cilia waft mucus up to throat to be swallowed                       |
| <b>Stomach acid (HCl)</b>         | Kills pathogens swallowed in food/water                                                    |
| <b>Tears</b>                      | Contain lysozyme — enzyme that kills bacteria                                              |
| <b>Inflammation / fever</b>       | Increased blood flow brings white blood cells; raised temperature inhibits pathogen growth |

## The immune system — white blood cells

### White Blood Cells in Action



**PHAGOCYTOSIS**  
non-specific  
(any pathogen)



**ANTIBODY PRODUCTION**  
specific to one antigen  
(memory cells stay)

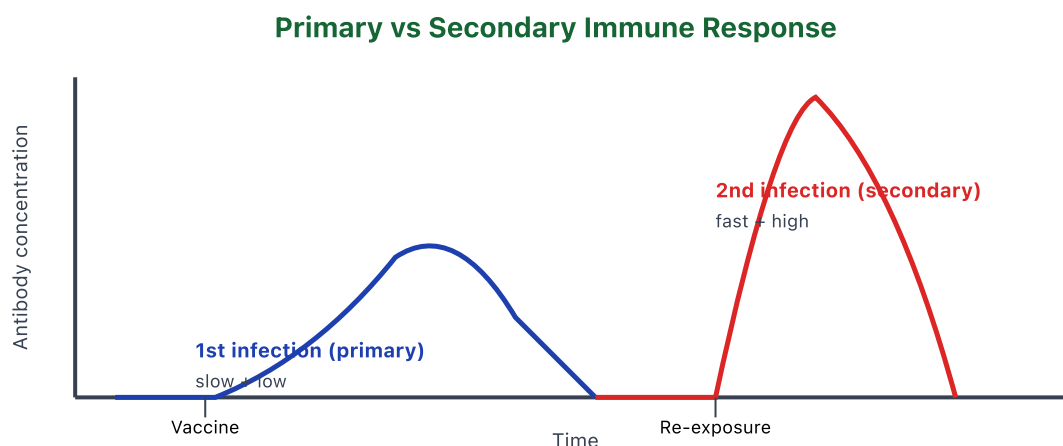
Some lymphocytes also release antitoxins

| Type of WBC       | Action                                                                                                                                                                                                |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phagocyte</b>  | Engulfs and digests pathogens (phagocytosis) — non-specific                                                                                                                                           |
| <b>Lymphocyte</b> | Produces antibodies that bind to specific antigens on the pathogen, marking them for destruction; also produces antitoxins to neutralise toxins; some become "memory cells" giving long-term immunity |

## 4. Vaccination

A **vaccine** contains dead or weakened pathogens (or pathogen fragments) that are introduced into the body. Lymphocytes produce antibodies and form **memory cells**, so if the real pathogen later infects the body, the immune response is **faster and stronger** — the person is **immune**.

## Primary vs secondary immune response



## Herd immunity

If a high enough proportion of the population is vaccinated, even unvaccinated people are protected because the disease cannot spread easily — this is **herd immunity**. For highly contagious diseases like measles, around 95% vaccination is needed.

## 5. Antibiotics and Antiviral Drugs

| Drug type         | Targets                                         | Examples                             |
|-------------------|-------------------------------------------------|--------------------------------------|
| <b>Antibiotic</b> | BACTERIA only — kill them or stop them dividing | Penicillin, amoxicillin, methicillin |
| <b>Antiviral</b>  | Viruses — slow their reproduction               | Used for HIV, herpes, flu            |
| <b>Antifungal</b> | Fungal infections (e.g. athlete's foot)         | Clotrimazole creams                  |

|                    |                                        |                                 |
|--------------------|----------------------------------------|---------------------------------|
| <b>Painkillers</b> | Treat SYMPTOMS — do not kill pathogens | Paracetamol, ibuprofen, aspirin |
|--------------------|----------------------------------------|---------------------------------|

**Antibiotics do NOT work on viruses** because viruses live inside host cells and use host machinery — there is no bacterial structure to target. Overuse causes antibiotic resistance (see B5.2).

## 6. Drug Development

New drugs are extensively tested before being released:

| Stage                                  | What's tested              | How                                                                                                          |
|----------------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>1. Pre-clinical</b>                 | Toxicity, dosage, efficacy | Tested on cells, then on tissues, then on animals                                                            |
| <b>2. Clinical trials Phase I</b>      | Safety; dosage             | Small group of healthy volunteers                                                                            |
| <b>3. Clinical trials Phase II/III</b> | Safety + efficacy          | Larger groups including patients with the condition; usually <b>double-blind</b> with <b>placebo</b> control |
| <b>4. Approval and monitoring</b>      | Long-term safety           | Drug released; effects in real population monitored                                                          |

**Placebo** = a "fake" treatment (e.g. sugar pill) used to control for the psychological effect of being treated. **Double-blind** = neither the patient nor the doctor knows who is receiving the real drug — prevents bias.

## 7. Monoclonal Antibodies

**Monoclonal antibodies (mAbs)** are identical antibodies produced from a clone of a single B-lymphocyte. They are highly **specific** — they bind to one type of antigen only.

## Production

1. A mouse is injected with a target antigen.
2. The mouse's lymphocytes produce antibodies.
3. The lymphocytes are extracted and fused with tumour cells (which divide rapidly) to make **hybridoma** cells.
4. Hybridoma cells divide indefinitely, producing huge quantities of identical antibodies.

## Uses

| Use                       | How                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------|
| <b>Pregnancy tests</b>    | mAbs detect the hormone hCG in urine — line on stick goes blue if pregnant               |
| <b>Diagnosing disease</b> | mAbs labelled with a fluorescent dye bind to specific tissues — show location of cancers |
| <b>Cancer treatment</b>   | mAbs can carry drug or radiation directly to cancer cells, sparing healthy cells         |
| <b>Research</b>           | Identifying specific molecules in tissues / cells                                        |

## 8. Non-Communicable Diseases

| Disease                             | Risk factors                                                                     |
|-------------------------------------|----------------------------------------------------------------------------------|
| <b>Cardiovascular disease (CVD)</b> | Smoking, high-fat diet, lack of exercise, high blood pressure, obesity, genetics |
| <b>Type 2 diabetes</b>              | Obesity, lack of exercise, poor diet, genetics                                   |
| <b>Lung cancer / disease</b>        | Smoking, pollution, asbestos                                                     |
| <b>Liver disease</b>                | Excess alcohol, obesity, hepatitis (viral)                                       |

|                     |                                                                                      |
|---------------------|--------------------------------------------------------------------------------------|
| <b>Many cancers</b> | Smoking, UV, radiation, alcohol, viruses (e.g. HPV → cervical cancer), age, genetics |
|---------------------|--------------------------------------------------------------------------------------|

## CVD treatments

- **Lifestyle changes:** stop smoking, healthier diet, more exercise — first line
- **Statins:** drugs to lower cholesterol
- **Stents:** mesh tubes to hold open narrowed coronary arteries
- **Bypass surgery:** replace blocked artery with a vein from the leg
- **Heart transplant:** in severe cases

## 9. Plant Diseases

| Disease                           | Cause     | Symptoms                                                         |
|-----------------------------------|-----------|------------------------------------------------------------------|
| <b>TMV (Tobacco Mosaic Virus)</b> | Virus     | Mosaic pattern on leaves; less photosynthesis → reduced growth   |
| <b>Rose black spot</b>            | Fungus    | Black spots on leaves; leaves drop early; reduced photosynthesis |
| <b>Crown gall</b>                 | Bacterium | Tumour-like growths on stems                                     |

## Plant defences

| Type              | Examples                                                                                   |
|-------------------|--------------------------------------------------------------------------------------------|
| <b>Physical</b>   | Cellulose cell walls, waxy cuticle, layers of dead cells (bark)                            |
| <b>Chemical</b>   | Antibacterial chemicals (e.g. mint), poisons (e.g. caffeine, nicotine)                     |
| <b>Mechanical</b> | Thorns, hairs, drooping leaves; mimicry (e.g. passion flower egg-like spots deter insects) |

**Mineral deficiencies in plants — see B6.2 table.**